Supplementary material for "What Do LLM Judges Actually Verify? Matched Controls and Record Gaps for Document Currency"

Unnati Shah
University of Southern California
ubshah@usc.edu

Samruddhi Kakade
Illinois Institute of Technology
samruddhikakade2001@gmail.com

Aum Patel
New York University
ap9018@nyu.edu

Supplementary material: the case study in full

This supplement holds the full report of the descriptive case study on Indian rooftop-solar regulation that the paper summarises in its final appendix: the reference data and their human audit, the conditions, damage and scoring, the planned analyses and their results, the readings fixed in advance, the prompts, the per-clause results, the annotation agreement, the deviations from the analysis plan and the data statement. The case study's answer key failed the quality checks planned for it, so nothing here supports a finding of the main paper; it is reported for transparency, with its planned analyses and readings kept as they were fixed. The sections are carried over from the study's own report, written before the RFC study, and use its terms (reviewer for judge, registry for record, clause for document).

S1 Case study: reference data, conditions and planned analyses

S1.1 Reference data and its audit

Corpus.

The corpus holds 111 authentic regulatory instruments on rooftop solar from 8 jurisdictions in India: central schemes and the regulations and orders of state commissions and distribution companies. Superseded instruments stay in the corpus, because their continued availability is the hazard studied. Every file is hashed and listed with its issuer, dates and source URL.

Amendment registry.

The reviewer's amendment record is a registry of 186 clause-level links, each from an old clause to the clause that changes it, with a relation (supersedes, partially

supersedes, amends), an effective date and a quote from the amending instrument. It was extracted from the documents by an AI assistant, without access to the gold graph used for scoring. It is the source of the R3 quotes, the R4 and R4n status lines and the R5 lookup. The registry as corrected by adjudication (below) is the one every run uses.

Cases and reference answers.

A case is a synthetic application: an electricity bill, an installer proposal and a distribution-company portal record, with a case date and a jurisdiction. Cases were generated backward from rule functions whose parameters are quoted from the clause text, so no label was typed by hand. Each case records the rule function that labelled it; most cases list a single applicable rule. The applications contain no personal data. The candidate test set had 60 cases on the replaced clauses with enough fixed answers per condition; the dev split shares no replaced clause with it and was used only for engineering and the pilot.

S1.1.1 Human audit of the reference data

The registry, the answer key and the case labels were produced with AI help, so two authors checked them blind before the test set was sealed, splitting the work and labelling a seeded shared subset (20 cases, 15 links) in common. They labelled each case's decision, blocking stage and deciding clauses from the case file and the source documents; each checked link's relation, effective date and wording; and a sample of 40 matched correct answers. A third author, who led the pipeline that produced the first-pass labels, adjudicated every disagreement from the annotators' sheets and the source documents, without the answer keys. The planned procedure had the annotators meet first; to save time the meeting was dropped and all disagreements went to the adjudicator (Appendix S6). The annotators are not legal professionals.

On the shared cases the annotators agreed on the decision in 80% of cases (kappa 0.60) and on the blocking stage in 85% (kappa 0.79), but on the exact set of deciding clauses in only 15% (kappa 0.14). On shared links they agreed on the relation in 87% (kappa 0.70). Part of the clause-level disagreement is notation (`cl-4.2` against `cl-4(2)`); part is that several clause sets support the same outcome.

The answer key.

Against the adjudicated labels, the AI-built key reached a different outcome on 18 of the 60 cases: 13 on the decision and 5 on the blocking stage of a rejection. In the disagreements we inspected, the readers based their decision on a second clause that the case's rule function does not evaluate, often in a newer instrument. In one such case, the key approves an application because it passes the transformer-capacity rule its rule function checks; the readers reject it under an eligibility clause of a later regulation. These cases are wrong at the base, and they are dropped; a case is never relabelled, because its faults were planted on the key's answer. On 36 of the remaining cases, the outcome agrees and only the deciding clauses differ. Those cases are kept and listed, because a planted fault and its catch are defined on the replaced clause

alone. Applied as originally planned, which dropped clause-only disputes too, the rules would have kept 6 cases.

The correct answers.

Of the 40 matched correct answers checked, 23 (57%) were judged correct. Most of the rest were judged incomplete or citing a wrong clause, the same single-rule cause: they cite the clause the rule function evaluated and omit another that also decides the case. A reviewer that flags such an answer may be right, so these items bias false alarms upwards. The planned gate (95% correct) would have stopped the seal; it was replaced by reporting, and by sensitivity set S1, which removes every case whose checked correct answer was judged anything but correct (§S1.3).

The registry.

The 52 checked links are the 18 that test items rely on and a seeded 20% random sample of 34 of the other links. In the random sample, 25 were confirmed, 4 had a different relation, and 5 were judged to be no change at all ([6, 30]%, Wilson 95% interval). Among the test-item links, 1 was no change and 3 had a different relation. Rejected links were removed and corrected dates applied before the run. These figures describe links the registry asserts. Nothing in the audit looked for links the registry misses, so we do not estimate that rate. The full table, with both annotators' raw answers beside the adjudicated one, is released (`40_data/link_audit_v1`); links outside the check keep the extractor's provisional relation and are not used as gold.

The scorer.

One annotator labelled 50 dev replies by hand under the written scoring rule. The automatic scorer agreed on 35 of 50 (70%), below the planned 95% gate. In each of the 15 disagreements, the reply's own clause ids and reason codes give the scorer's label under the written rule, so the scorer was not changed. Most disagreements lie on the boundary between a wrong clause and a wrong reason (10), or between a right catch and a wrong reason (3). We read this as evidence that the right-reason rule is hard to apply by hand, not as a defect of the scorer, and report every result under the two looser rules as well (E5).

The sealed test set.

The sealed set has 42 cases on 14 distinct replaced clauses from 6 jurisdictions (Table S1). 3 clauses in one state were replaced by a single amending instrument and are not independent. The registry links 11 of the 14 replaced clauses; on the others the intact registry itself reads `in_force`, so they are naturally missing links.

S1.2 Conditions, damage and scoring

S1.2.1 Fixed answers

We measure review on fixed answers, so the result does not depend on what a generator happens to write. For each case and condition we build a matched pair. The *fault* cites

Condition	Faults	Matched correct answers	Clauses
S-both (old and current clause shown)	42	42	14
S-swap (old clause in place of current)	38	7	13
C (clean correct answers)	–	42	–

Table S1 Fixed answers per condition in the sealed test set. Clauses: distinct replaced clauses. Jurisdictions of the replaced clauses: DL 4, GJ 4, central 1, KA 2, TN 1, UP 2.

the replaced clause in place of its successor. When the old clause changes the approve-or-reject outcome, the fault also carries the decision that clause implies; otherwise it has the right decision on an invalid basis. The *matched correct answer* (control) has the key's decision and cites only clauses its own evidence shows. Both come from one neutral template: the decision, the cited clause identifiers, one sentence that quotes the basis clause, and the blocking stage if any. A pair therefore differs only in its basis and, where the outcome changes, in the decision. Condition C adds one clean correct answer per case. The reviewer sees the case, the evidence and one answer, and is never told which answers are faults.

The evidence differs by condition. In *S-both* the old clause and its successor are both shown; in *S-swap* the old clause replaces the current one, so the reviewer sees only outdated text.

S1.2.2 Conditions

Every language-model condition uses the same validity prompt, which asks whether each cited clause is in force on the application date, issued for the case's jurisdiction and applicable to its facts. Conditions differ only in what the prompt contains (Table S2; prompts in Appendix S3). R3, R4 and R4n are built on R2: the R3 block holds the amending instrument's verbatim substitution text for each cited clause the registry links, with no status, date or target; the R4 block holds one line per cited clause giving the registry's entries, jurisdiction and status at the case date. R4′ gives the same line at R1, without the current clause; it shows what the status line does when the text cannot confirm it. R5 uses no language model: it raises a "not in force" finding on a cited clause exactly when the registry marks that clause superseded or partially superseded at the case date (an "amends" link alone does not), and it raises no other kind of finding. Its catches are therefore the registry's own links, and its false alarms are the registry's own wrong links.

The no-record arm.

In the registry, a clause with no outgoing link has status `in_force`. The registry cannot distinguish a clause that was never amended from one whose amendment it failed to record, so this value states more than the record holds. R4n replaces the status value `in_force` by `no_amendment_recorded` and changes nothing else: the prompt sent at R4n equals the R4 prompt with that one substitution, which a unit test checks on an intact and a damaged registry. Superseded, partially superseded and not-yet-in-force values are unchanged. The substitution applies to every clause the registry does not

Condition	Reviewer sees	Reviewer does not see	Compared with; confound	Role
R0	old clause (S-swap), source metadata removed	current clause, dates, issuer	–	ladder
R1	old clause with issuer, dates, title	current clause	R0: metadata only	ladder
R2	old and current clause (S-both)	any amendment record	R1: adds the current clause	ladder; Q2 baseline
R3	R2 + amending instrument's words	status, dates, target	R2	ladder
R4	R2 + registry status line	–	intact: told the answer for linked clauses	reference
R4′	R1 + registry status line	current clause	R1: status line without the text	ladder
R4-damage	R2 + status line of a damaged registry	which links are wrong	R5-damage (Q1); R2 (Q2)	primary
R4n, R4n-damage	as R4, "no amendment recorded" for unlinked clauses	–	R4 on the same items and seeds	primary
R5, R5-damage	no LLM: "not in force" on every cited clause the registry marks superseded	–	catches exactly the linked clauses, by construction	reference

Table S2 Conditions. "Reference" rows are not findings: R5 recovers the registry's own links, and R4 at the intact registry is told, for every linked clause, that it is not in force. Faults were planted on replaced clauses of the gold graph; the registry links most but not all of them (§S1.1). The E2 variants of R0 to R2 add a "can't confirm" reason.

link, including current clauses on correct answers, so R4n can change false alarms as well as catches.

Can't confirm.

For E2, variants of R0, R1 and R2 add the reason `currency_unconfirmed`: the evidence shown does not establish whether the clause is in force at the case date. Such a finding is neither a catch nor a false alarm.

S1.2.3 Damaged registries

To vary the record's reliability in a controlled way we damage the registry in three ways: 25% or 50% of its links dropped, or 10% spurious links added. Each damage has 5 seeds. A drop permutes all of the registry's links with the seed and removes the first quarter or half, so at one seed the 50% set contains the 25% set. A spurious link is added from a seeded choice of dated clauses that have no outgoing link, to another clause of the same jurisdiction, as a "superseded by" link effective from the clause's own date; the number added is 10% of the registry's links. A damaged registry is used in each way the intact one is: as the lookup (R5-damage) and as the status line (R4-damage, R4n-damage). A dropped link makes the status line report a replaced clause as `in_force` (R4) or as having no recorded amendment (R4n), while the evidence still

shows both versions; a spurious link makes it report a current clause as replaced. In both cases the text in view contradicts the line. The damage is random and synthetic; it is not an estimate of how real registries fail.

S1.2.4 Scoring

The reviewer returns a verdict and findings; each finding names a clause and gives one reason from a closed list (not in force, wrong jurisdiction, not applicable, missing rule, content or arithmetic, other). The list is the same in every prompt and names no clause.

Catch, false alarm, confusion counts.

A fault is caught when a finding names the replaced clause with the reason "not in force"; an identifier of a retrieval chunk counts as the clauses it contains. On a correct answer, any finding is a false alarm. Each condition therefore yields four counts: caught and missed faults, flagged and passed correct answers. A finding on the right clause for another reason is not a catch; we count it as a *misnamed* alarm (E4).

Discrimination.

For one replaced clause, discrimination is its catch rate on faults minus the false-alarm rate on their matched correct answers; it is averaged over clauses, and for a damaged registry first over seeds within a clause. It is the headline measure, because a reviewer that flags every answer catches every fault and is of no use.

Deference and NoRec.

Both use the same items: every (S-both fault, drop rate, seed) where the intact registry catches the fault and that damaged registry does not. Deference is, item by item, the reviewer's catch with no registry (R2) minus its catch with the damaged status line (R4-damage). NoRec is the item's catch at R4n-damage minus its catch at R4-damage on the same seed. Both are averaged within clause, then over clauses.

Three scoring rules (E5).

Under *any finding*, a fault is caught if the reply has any finding; under *clause named*, if a finding names the replaced clause for any reason; the right-reason rule above is the strictest. On correct answers we report any finding (our false-alarm rule) and a finding with a validity reason only.

Post hoc measures (added in review, not pre-specified).

Because the lookup can only raise "not in force" while the reviewer can raise any reason, we also report Q1 with false alarms counted only when the finding gives a validity reason (not in force, wrong jurisdiction, not applicable), the reason codes behind the false alarms, the catches the reviewer gains and loses against the lookup on the same item and seed, and Q1 with the amending instrument rather than the clause as the unit (the 14 clauses come from fewer amending instruments). These are descriptive and outside the family.

Invalid replies.

A reply that is not valid JSON, is cut off at the output cap after one retry, or has no verdict is invalid. It counts as a miss on a fault and as a false alarm on a correct answer, so an invalid reply never improves a condition.

S1.3 Setup and planned analyses

Models.

The panel has 4 open models: `llama3.1:8b` (weak), `qwen2.5:7b`, and the 2 capable models `qwen2.5:14b` and `gemma-4-26b-a4b-it`. The first three run through Ollama with 4-bit quantised weights fixed in advance by digest (`46e0c10c039e`, `845dbda0ea48` and `7cdf5a0187d5`; full digests in the analysis plan), checked on every machine: `qwen2.5:14b` on a cloud notebook with two NVIDIA T4 GPUs, the other two on an Apple-silicon laptop. Gemma, open-weight, runs through a hosted API, with the version recorded per call. A closed hosted model, `gemini-3.5-flash-lite`, is a secondary model on R0, R2 and R4 only. Each model ran on one machine. Models added after filing are exploratory and reported separately (§S1.4.5).

Settings.

Decoding is greedy (temperature 0), with run seed 42 and an output cap of 1024 tokens. Every reply is cached, and every block records the model digest, the prompt hash per item, the code hash and the hash of the sealed test file. Every model sees the same items.

Run order and cutoff.

Blocks run in a fixed priority order: for the capable models R2, R4, R4n, R5 and R5-damage, then R4-damage and R4n-damage seed by seed, so that a cut run still leaves the two arms paired on the same seeds; then the secondary conditions; then the other models. The analysis cutoff was fixed as a wall-clock time before any test result was read. A block not finished by then is reported as not run, and a contrast whose blocks are incomplete is reported as not run rather than estimated from part of a block (Appendix S6 lists what did not run).

Unit of inference and tests.

The unit of inference is the distinct replaced clause. Rates are averaged within each clause and then over clauses, with pooled counts beside each rate; false alarms on condition C are averaged by rule group, because C plants nothing. A contrast is the mean per-clause difference, paired by clause within one model, tested with a two-sided sign-flip test over clauses (exact for few clauses, otherwise 10,000 random sign vectors with a fixed seed). Intervals are 95% bootstrap intervals over clauses, printed only with at least 10 clauses. A cell with fewer than 5 items or 3 clauses is printed as counts.

Primary family.

10 tests, Holm-corrected at $\alpha = 0.05$: for each capable model, R4 minus R5 discrimination at the intact, 50%-drop and 10%-spurious registries (Q1), deference (Q2) and

NoRec. A contrast on fewer than 10 clauses enters the family with $p = 1$. The 25% drop is reported beside Q1, uncorrected. Everything else is secondary and uncorrected or descriptive: the ladder contrasts (E1), can't confirm (E2), spurious-link false alarms (E3′), misnamed alarms (E4), the scoring rules (E5), the false-alarm cost of R4n, and descriptive splits by amendment wording (explicit or implicit), by whether the fault changes the outcome, and by cost per call (tokens and seconds).

Sensitivity sets.

Every primary result is also computed on two subsets, with the same tests and outside the family. **S1** removes every case whose checked correct answer was judged anything but correct. **S2** removes every case the originally planned rule would have dropped, that is, every case whose deciding clauses differ from the key; it has fewer than 10 clauses and is descriptive. Only a sample of correct answers was checked, so S1 removes the correct answers known to be flawed, not every flawed one; a result that keeps its sign under S1 does not depend on those.

Damage seeds.

The capable models run R4-damage and R4n-damage on all 5 seeds; the other models run seed 0. R5-damage is deterministic and runs every seed.

Analysis plan.

The design, outcomes, analysis and the sentence reporting each outcome were written into a dated, version-controlled analysis plan before any model saw a test item, and the test set was sealed with a fingerprint recorded in the repository. The plan was not filed with a public registry; its history is in the released version control. Two dated amendments were made before the seal. The first, written after the dev pilot, made the damaged-registry comparisons primary. The second, written after the human audit and before any model saw a test item, changed which cases annotation keeps, replaced the correct-answer and scorer gates by reporting, and added R4n, NoRec, S1 and S2 (Appendix S6).

S1.4 Planned results

Seven reviewers ran on the same items: the five planned models, and two added after the analysis plan was fixed, `qwen2.5:32b` (damage seeds 0–2) and Claude Opus 4.5 through a hosted proxy (damage seed 0; temperature and seed set by the proxy). Every planned block finished for the planned models. Table S3 gives the confusion counts of every main arm on the same 42 S-both faults and 42 matched correct answers. A repeat of `qwen2.5:14b`'s R2, R4 and R4n blocks on the same machine with an empty cache returned identical findings on 378 of 378 items; we did not test whether replies are identical across machines or for the hosted models. Invalid replies were rare (Gemma: 5/630 at R4 with 50% of links dropped; `qwen2.5:14b`: none).

Table S3 Main results on S-both faults and their matched correct answers. TP: right-reason catches of the planted superseded clause; FN: faults not caught; FP: matched correct answers with any finding; TN: matched correct answers passed. Recall, specificity, precision and balanced accuracy (%) pool items, which share clauses, so they are descriptive. FA_C: clean correct answers flagged. Disc.: discrimination (catch minus matched false alarm), clause-averaged, with a 95% bootstrap interval over clauses (points). R5 recovers the registry's own links by construction and R4 (intact) is given the answer: reference rows.

Model	Arm	Reviewer sees	TP	FN	FP	TN	Rec.	Spec.	Prec.	Bal.	FA_C	Disc. [95% CI]	Cl.
Qwen 14B	R2	both versions	13	29	18	24	31	57	42	44	12/42	−7 [−38, +20]	14
Qwen 14B	R4	status line	32	10	10	32	76	76	76	76	11/42	+55 [+26, +79]	14
Qwen 14B	R4n	no-record wording	31	11	13	29	74	69	70	71	12/42	+47 [+17, +73]	14
Lookup	R5	lookup (ref.)	31	11	5	37	74	88	86	81	5/42	+67 [+43, +88]	14
Qwen 14B	R4-drop50	status, drop 50%	88	122	54	156	42	74	62	58	55/210	+21 [−6, +48]	14
Qwen 14B	R4n-drop50	no-record, drop 50%	95	115	58	152	45	72	62	59	60/210	+23 [−6, +54]	14
Lookup	R5-drop50	lookup, drop 50%	88	122	8	202	42	96	92	69	8/210	+42 [+24, +60]	14
Qwen 14B	R4-spurious10	status, spur. 10%	160	50	55	155	76	74	74	75	61/210	+53 [+25, +76]	14
Lookup	R5-spurious10	lookup, spur. 10%	155	55	34	176	74	84	82	79	34/210	+62 [+37, +84]	14
Gemma 4 26B	R2	both versions	9	33	20	22	21	52	31	37	17/42	−20 [−46, +4]	14
Gemma 4 26B	R4	status line	25	17	20	22	60	52	56	56	18/42	+21 [−18, +57]	14
Gemma 4 26B	R4n	no-record wording	26	16	21	21	62	50	55	56	15/42	+18 [−20, +54]	14
Gemma 4 26B	R4-drop50	status, drop 50%	89	121	100	110	42	52	47	47	90/210	+4 [−30, +35]	14
Gemma 4 26B	R4n-drop50	no-record, drop 50%	96	114	105	105	46	50	48	48	72/210	+2 [−31, +31]	14
Gemma 4 26B	R4-spurious10	status, spur. 10%	125	85	103	107	60	51	55	55	94/210	+20 [−20, +56]	14

	`qwen2.5:14b`			`gemma-4-26b`		
Registry	R4	R5	R4−R5	R4	R5	R4−R5
Intact	+55	+67	−11	+21	+67	−45
Drop 25%	+39	+51	−12	+14	+51	−37
Drop 50%	+21	+42	−21	+4	+42	−38
Spurious 10%	+53	+62	−10	+20	+62	−42

Table S4 Discrimination in points of the LLM reviewer with the status line (R4) and of the registry lookup alone (R5), by registry state, over 14 clauses. The difference is the Q1 contrast; intervals and Holm p are in Table S5. The 25% row is outside the primary family.

Table S5 Primary contrasts (capable models; Holm over the family of ten). Per-clause differences in points, mean over clauses, 95% bootstrap interval over clauses, two-sided sign-flip test. S1: without the cases whose checked correct answer was judged flawed; S2: under the original strict annotation rule (descriptive when it leaves fewer than ten clauses).

Model	Contrast	Clauses	Diff.	95% CI	p (Holm)	S1	S2
Gemma 4 26B	Q1, intact: R4 − R5 discrimination	14	−45	[−69, −24]	0.043	−47	−50
Gemma 4 26B	Q1, 50% dropped	14	−38	[−65, −11]	0.178	−38	−50
Gemma 4 26B	Q1, 10% spurious	14	−42	[−66, −22]	0.043	−45	−45
Gemma 4 26B	Q2: R2 − R4-damage catch, dropped links	9	+1	–	1.000	+4	–
Gemma 4 26B	R4n − R4 catch, dropped links	9	+9	–	1.000	+20	–
Qwen 14B	Q1, intact: R4 − R5 discrimination	14	−11	[−33, +9]	1.000	−12	+17
Qwen 14B	Q1, 50% dropped	14	−21	[−38, −6]	0.206	−17	+2
Qwen 14B	Q1, 10% spurious	14	−10	[−30, +10]	1.000	−11	+17
Qwen 14B	Q2: R2 − R4-damage catch, dropped links	9	+34	–	1.000	+33	–
Qwen 14B	R4n − R4 catch, dropped links	9	+9	–	1.000	+9	–

S1.4.1 Q1: does the reviewer add anything over the record?

Under the planned rule, the reviewer given the record had a lower point estimate of discrimination than a lookup in the same record at every registry state, for both capable models (Table S4). For Gemma the difference was −45 points with the registry intact (CI [−69, −24], Holm $p = 0.043$) and −42 with spurious links (CI [−66, −22], Holm $p = 0.043$), both significant after Holm, and −38 with half the links dropped

(Holm $p = 0.178$). For `qwen2.5:14b` it was -11 (Holm $p = 1.000$), -21 (CI [-38, -6], Holm $p = 0.206$) and -10 (Holm $p = 1.000$), none significant. The reading fixed in advance for both models is that the reviewer did not discriminate better than a lookup in the registry it was given, even when the registry was damaged; the reading fixed in advance for Gemma's intact result is that adding the reviewer to a correct registry made review worse than the registry alone.

That last reading should be taken narrowly, for three reasons we found after the run (post hoc, §S1.2.4). First, the gap is made of findings the lookup cannot raise. The planned false-alarm rule counts any finding on a correct answer, but the lookup only ever says "not in force". Most of Gemma's findings on matched correct answers gave "content or arithmetic" or "missing rule" as the reason, not a validity reason. Counting only validity-reason findings as false alarms, the difference is -6 points for Gemma with the registry intact (CI [-20, $+8$]) and $+1$ with half the links dropped; for `qwen2.5:14b` it is -3 and -13 (CI [-28, 0]). On the question the lookup answers, whether a cited clause is in force, the reviewer did about as well as the lookup. Its other findings may be right: the audit found that many correct answers omit a clause that also decides the case (§S1.1.1), which this design cannot score. Second, the 14 clauses come from 8 amending instruments; with the instrument as the unit, Gemma's intact and spurious differences have uncorrected sign-flip $p = 0.031$ and 0.031, on fewer units than the planned minimum of 10. Third, both lose significance under S1 (§S1.4.6). We therefore report Gemma's Q1 result as planned, but do not read it as evidence that a reviewer degrades a correct record.

The expectation fixed in advance for Q1, that the gap would narrow or reverse as links were dropped because the reviewer can still read both versions, was not met. Table S3 shows why. With 50% of links dropped, the reviewer's right-reason catches on S-both faults nearly equal the lookup's for both models. Item by item and seed by seed, Gemma caught 25 faults the damaged lookup missed and missed 24 it caught; `qwen2.5:14b` gained 13 and lost 13. The reviewer recovered some dropped links from the text, and lost about as many catches elsewhere, so on net it did not repair the record. Its false alarms on matched correct answers stayed well above the lookup's.

Two properties of the design bound this result. The lookup's catches are the registry's own links: it misses the faults on clauses the registry does not link and flags whatever a wrong link marks, so its discrimination is a property of the record, not a benchmark of reasoning (§S1.2). And the result holds for these models, these 14 clauses and fixed answers with a single planted fault.

S1.4.2 Q2: does the reviewer follow a wrong "in force" line?

The deference contrast could be computed on 9 clauses: those whose planted link the intact registry catches and some drop seed removes. That is below our planned minimum of 10, so by rule it entered the family with $p = 1$ and supports no confirmatory claim; we report it descriptively and, as planned, as no evidence of deference rather than evidence of its absence. On those faults `qwen2.5:14b` caught 41% with no registry (R2) and 7% when the damaged status line said the replaced clause was in force (deference +34 points). Gemma caught 32% and 31% (deference +1). The direction for `qwen2.5:14b` is the one the deference account predicts and matches the effect of

		qwen2.5:14b			gemma-4-26b		
	Rule	R0	R2	R4	R0	R2	R4
Catch	any finding	40%	62%	79%	50%	76%	79%
	clause named	40%	62%	79%	48%	76%	79%
	right reason	8%	33%	79%	0%	24%	64%
False alarm	any finding	40%	40%	23%	40%	45%	43%
	validity reason	40%	23%	15%	20%	0%	4%

Table S6 The same replies under three scoring rules, on faults (catch) and matched correct answers (false alarm). Clause-averaged. The right-reason row is the catch reported everywhere else. At R0 the reviewer sees only the old clause, so a right-reason catch is not expected there.

false assurances that Liu and Gao (2026) report for code auditors; Gemma's catches on these items hardly changed.

The no-record wording (R4n).

On the same faults, stating that no amendment is recorded instead of that the clause is in force raised catches from 7% to 16% for `qwen2.5:14b` and from 31% to 41% for Gemma (NoRec +9 and +9 points, 9 clauses, below the minimum, no confirmatory claim). False alarms on matched correct answers rose a little: with half the links dropped, from 25% to 26% for `qwen2.5:14b` and from 43% to 48% for Gemma; with the registry intact, from 23% to 30% and from 43% to 48%. As planned, we report no evidence that the wording matters, not evidence that it does not; the point estimates are small and positive, recover a fraction of what the dropped links cost, and come with a small rise in false alarms. The set is small by construction: deference and NoRec need a fault the intact registry catches, and the registry links only 11 of the 14 clauses.

Spurious links (E3′).

On correct answers citing a clause that a spurious link marks as replaced (9 items on 3 clauses), `qwen2.5:14b` raised a false alarm on 9/9 with the status line against 6/9 with no registry; Gemma on 8/9 against 4/9. A wrong link marking a current clause as replaced was followed into a false alarm more often than not. The counts are small and no test was planned. Unlike the code auditors of Liu and Gao (2026), whose accuracy was never significantly lowered when a tool raised false alarms, these reviewers followed a wrong record in both directions; on three clauses this is a hypothesis for confirmatory work, not a finding.

S1.4.3 How much the scoring rule matters

The scoring rule changed the apparent skill of the same replies by tens of points (Table S6). With both versions in view (R2), any-finding scoring credited `qwen2.5:14b` with 62% catches and Gemma with 76%, against 33% and 24% when the finding had to name the replaced clause for the reason that it is no longer in force. Naming the clause hardly mattered: almost every finding on a fault named it, and the gap is the reason. Under any of the rules the reviewers also flagged a large share of matched correct answers, so a catch rate reported alone would overstate them. The reading

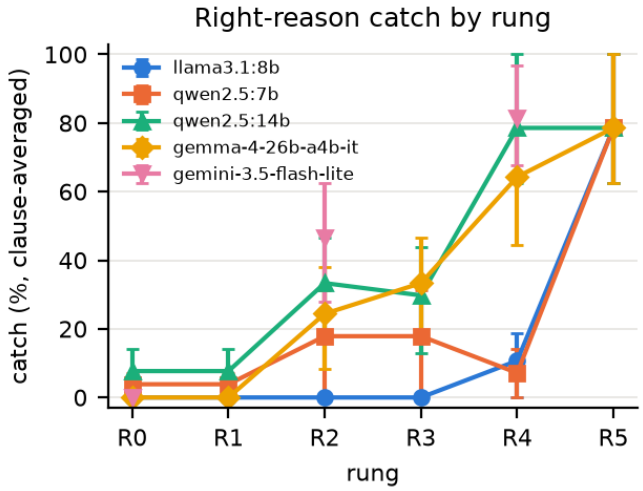


Fig. S1 Right-reason catch by rung, one line per model, clause-averaged, with 95% bootstrap intervals over clauses. R5 is the registry lookup and is the same for every model.

	llama3.1:8b			qwen2.5:7b			qwen2.5:14b			gemma-4-26b		
Rung	Catch	FA-m	FA-C	Catch	FA-m	FA-C	Catch	FA-m	FA-C	Catch	FA-m	FA-C
R0	0%	80%	51%	4%	80%	20%	8%	40%	26%	0%	40%	34%
R1	0%	80%	49%	4%	60%	12%	8%	50%	27%	0%	20%	36%
R2	0%	49%	49%	18%	18%	12%	33%	40%	27%	24%	45%	36%
R3	0%	49%	49%	18%	20%	19%	30%	36%	27%	33%	45%	36%
R4	11%	49%	51%	7%	7%	5%	79%	23%	26%	64%	43%	39%
R4′	6%	80%	51%	6%	40%	5%	81%	0%	26%	62%	40%	39%
R5	79%	13%	12%	79%	13%	12%	79%	13%	12%	79%	13%	12%

Table S7 Right-reason catch and false alarms by rung (clause-averaged; C false alarms averaged by rule group). FA-m: matched correct answers of the same blocks (S-swap pairs for R0, R1 and R4′, S-both otherwise; R5 pools both). FA-C: clean correct answers. Table S3 gives S-both counts only, so its false alarms can differ from FA-m here.

fixed in advance applies: scoring by flags alone would have credited these reviewers with many catches made for the wrong reason.

The right-reason rule is itself hard to apply. In the human check of the scorer on 50 dev replies the annotator and the code agreed on 35 of 50 (70%), below the pre-specified 95% gate; 10 of the 15 disagreements lie on the boundary between a wrong clause and a wrong reason. Amendment 2 kept the scorer and reports the failed gate (§S1.1.1); the any-finding and clause-named rows of Table S6 show every result under the looser rules, and the post hoc Q1 analysis in §S1.4.1 applies the validity-reason rule to false alarms.

S1.4.4 The reference ladder (E1) and the other models

The ladder is the reference that places Q1 and Q2. For both capable models the status line lifted right-reason catches over seeing both versions: R4−R2 was +45 points for `qwen2.5:14b` (CI [+24, +65]) and +40 for Gemma (CI [+20, +58]), while R2−R0 was +28 and +26, not significant by the planned sign-flip test (Gemma's bootstrap interval, [+8, +48], excludes zero; R0 runs on S-swap items, where a right-reason catch is near zero by design). In the words fixed in advance, seeing both versions is not enough, and R4 lifts these capable models. The amending instrument's quote (R3) added little over R2 (−4 and +9 points). Without a record the reviewers did not discriminate at all: R2 discrimination was −7 points for `qwen2.5:14b` and −20 for Gemma, because they flagged matched correct answers as often as, or more often than, they caught faults for the right reason.

The weak model barely moved: `llama3.1:8b` caught 0% at R2 and 11% at R4. In the words fixed in advance, below a capability threshold the information does not help; this is reported as a limit of the claim. `gemini-3.5-flash-lite`, run at R0, R2 and R4 only, caught 46% at R2 and 82% at R4, with false alarms on matched correct answers of 44% at R4; seeing both versions did not close the gap for it either, so stop rule 5 did not trigger. Given the "can't confirm" option (E2), neither capable model used it on a fault at R2 (0/42 and 0/42): the option was available and unused. Of the R2 findings that flagged the replaced clause, 13/26 (`qwen2.5:14b`) and 23/32 (Gemma) never gave "not in force" as the reason (E4).

S1.4.5 The added models

The two models added after the analysis plan was fixed are outside every planned test; Table S8 and the descriptive-results table of the paper's case-study appendix report them on the main arms. Claude Opus 4.5 is the clearest case that capability changes the picture: it is the only reviewer whose catches survive the absence of a record, and the only one that repaired a damaged record. A model from the same developer also helped build this benchmark (see the paper's statement on the use of AI assistants); its items were not tuned on any model's replies, but readers should weigh that relation. It ran one damage seed, through a proxy that sets temperature and seed itself, so its numbers carry more uncertainty than the planned models'. Scale did not buy a better reviewer. `qwen2.5:32b`, run with the pre-specified prompts on the same items, flagged matched correct answers more often than `qwen2.5:14b` at every arm, and about twice as often once given the record, so its discrimination was lower, and its gap to the lookup wider, with the record intact and with half the links dropped. Without a record it named the replaced clause on most faults but rarely for the reason that it is no longer in force. On the dropped-link faults its catches fell to none under the damaged status line, and the no-record wording did not restore them. One model, one family and the same 14 clauses cannot separate size from other differences between checkpoints; we report it because a reader will ask whether a larger model removes the pattern, and here it did not.

Table S8 Size comparison within one family. Rows marked "no" were added after the analysis plan was fixed and are exploratory; the planned sizes are shown beside them. The 32B model ran damage seeds 0–2 (the planned capable models ran 0–4). S-both right-reason catches, false alarms on matched correct answers (any finding) and any-finding catches, pooled k/n; discrimination clause-averaged (points), and the Q1 difference with a 95% bootstrap interval over clauses. No test.

Model	Planned	Arm	Catch	FA	Any	Disc. / diff.
Qwen 7B	yes	R2	5/42	7/42	14/42	−1
Qwen 7B	yes	R4	2/42	2/42	3/42	+0
Qwen 7B	yes	R4-drop50	2/42	2/42	3/42	+0
Qwen 7B	yes	R4−R5, intact				−67 [−88, −43]
Qwen 7B	yes	R4−R5, drop50				−42 [−60, −24]
Qwen 14B	yes	R2	13/42	18/42	26/42	−7
Qwen 14B	yes	R4	32/42	10/42	32/42	+55
Qwen 14B	yes	R4n	31/42	13/42	32/42	+47
Qwen 14B	yes	R4-drop50	88/210	54/210	126/210	+21
Qwen 14B	yes	R4−R5, intact				−11 [−33, +9]
Qwen 14B	yes	R4−R5, drop50				−21 [−38, −6]
Qwen 32B	no	R2	4/42	25/42	35/42	−48
Qwen 32B	no	R4	29/42	23/42	38/42	+18
Qwen 32B	no	R4n	26/42	24/42	38/42	+11
Qwen 32B	no	R4-drop50	46/126	69/126	104/126	−15
Qwen 32B	no	R4−R5, intact				−48 [−68, −24]
Qwen 32B	no	R4−R5, drop50				−57 [−76, −37]
Claude Opus 4.5	no	R2	34/42	40/42	42/42	−11
Claude Opus 4.5	no	R4	35/42	37/42	42/42	−3
Claude Opus 4.5	no	R4n	35/42	42/42	42/42	−14
Claude Opus 4.5	no	R4-drop50	29/42	37/42	42/42	−21
Claude Opus 4.5	no	R4−R5, intact				−70 [−89, −48]
Claude Opus 4.5	no	R4−R5, drop50				−63 [−85, −43]

S1.4.6 Sensitivity to the annotation audit

Removing the cases whose checked correct answer annotators judged flawed (S1; only a sample was checked) left the sign of every primary contrast unchanged (Table S5). Gemma's two significant Q1 differences became −47 and −45 points with Holm $p = 0.083$ and 0.083, no longer below 0.05 on the smaller set. Under the reading fixed in advance, the Q1 result does not depend on the correct answers known to be flawed in direction; its significance for Gemma does depend on keeping them. Under the original strict rule (S2) only 6 cases remain, on fewer than 10 clauses; that is descriptive only, and there `qwen2.5:14b`'s Q1 difference turns positive (+17 points intact), Gemma's stays negative (−50).

S2 Case study: readings fixed in advance

Each rule was planned with the sentence we report when it applies. Readings 1–5 are from amendment 1, readings 6–8 from amendment 2, and the rest from the original registration, where they now apply to the reference ladder.

1. R4−R5 not positive after Holm at any registry state, for a capable model: "For this model, the reviewer did not discriminate better than a lookup in the registry it was given, even when the registry was damaged."

2. R4−R5 positive after Holm at a damaged state but not with the registry intact: "The reviewer added value over the registry only once the registry was damaged, from [the first such state]."
3. R4−R5 negative after Holm with the registry intact: "Adding the reviewer to a correct registry made review worse than the registry alone."
4. Deference positive after Holm: "A wrong 'in force' line removed catches this reviewer makes without any registry: it followed the registry over the text." Not significant: "We found no evidence of deference"; never "the reviewer does not defer".
5. Any-finding catch above right-reason catch by more than 10 points for a capable model at R0 to R2: "Scoring by flags alone would have credited this reviewer with [the gap] points of catches it made for the wrong reason."
6. NoRec positive after Holm and the false-alarm cost small: "Telling the reviewer that no amendment is recorded, instead of that the clause is in force, recovers catches the damaged record loses." Not significant: "We found no evidence that the wording matters"; never that it does not.
7. NoRec positive but the false-alarm cost larger than the gain: "The wording trades catches for false alarms" (both numbers reported).
8. A primary result that keeps its sign under S1: "It does not depend on the flawed correct answers." One that changes sign: "It depends on them."
9. Fewer than 10 distinct replaced clauses in the sealed test: no primary claim. "With this many distinct clauses the test cannot support a population claim; we report per-clause results descriptively."
10. R2 at least as high as R4 for the capable models: "The bottleneck is retrieval, not representation: given both versions, reviewers catch the replaced rule as well as when told its status."
11. R4−R2 not significant (uncorrected, secondary since D48): no ladder claim. "Telling the reviewer the registry status did not reliably raise catches for these models."
12. The weak model at R4 within 10 points of its R2: "Below a capability threshold, the information does not help." Reported as a limit of the claim.
13. The secondary model's R2 within 10 points of its R4: "For a stronger model, seeing both versions is enough; the ladder result is limited to the local panel." Reported in the abstract.
14. "Can't confirm" never used by a model: "The option was available and unused."

"Seeing both versions is not enough" means that R2−R0 is not significant and that R2's catch is below half of R4's.

S3 Case study: prompts

The exact prompt of every rung, rendered for one dev item by `render_prereg_prompts.py`, is filed with the analysis plan (`PREREGISTRATION_v3_PROMPTS.md`) and released with the code; we do not retype it here. R4n is R4's prompt with one change: in the registry lookup block, a status value `in_force` reads `no_amendment_recorded`. A unit test checks that the two prompts differ in that word only, with the registry intact and damaged.

Table S9 Right-reason catches per replaced clause (k/n faults; R5 also counts S-swap faults), capable models. The clause is the unit of every interval and test in the paper.

Replaced clause	Qwen 14B				Gemma 4 26B			
	R2	R4	R4n	R5	R2	R4	R4n	R5
DL-C3-004#cl-3.8	2/4	4/4	4/4	8/8	0/4	4/4	4/4	8/8
DL-C3-004#cl-9.2b	2/2	2/2	2/2	4/4	2/2	2/2	2/2	4/4
DL-C3-006#cl-3.2	2/3	3/3	3/3	6/6	3/3	3/3	3/3	6/6
DL-C3-009#cl-3.6	2/4	4/4	4/4	8/8	1/4	4/4	4/4	8/8
GJ-C3-001#cl-13.5-check	0/4	4/4	4/4	8/8	0/4	2/4	3/4	8/8
GJ-C3-001#cl-13.5-class	3/3	3/3	3/3	6/6	2/3	3/3	3/3	6/6
GJ-C3-001#cl-5-proviso	0/3	0/3	0/3	6/6	0/3	0/3	0/3	6/6
GJ-C3-003#cl-6.2	0/2	2/2	2/2	4/4	0/2	2/2	2/2	4/4
IN-C1-011#cl-annex-phase2-claims	0/2	2/2	2/2	4/4	0/2	2/2	2/2	4/4
KA-C5-001#cl-15	0/4	4/4	3/4	0/8	0/4	0/4	0/4	0/8
KA-C5-001#cl-26	0/4	0/4	0/4	0/4	0/4	0/4	0/4	0/4
TN-C5-001#cl-2	0/3	0/3	0/3	0/6	0/3	0/3	0/3	0/6
UP-C3-001#cl-5.1	1/2	2/2	2/2	4/4	0/2	1/2	1/2	4/4
UP-C3-001#cl-6.2	1/2	2/2	2/2	4/4	1/2	2/2	2/2	4/4

S4 Case study: per-clause results

Every clause-level rate in the paper is built from these cells; the released `ladder_per_clause.csv` has them for every model and rung, with controls, control alarms, can't-confirm, misnamed and invalid counts.

S5 Case study: annotation agreement

On the seeded shared subset (15 links, 20 cases) labelled by both annotators:

Sheet	Field	Agreement	κ	PABAK
Links	relation	87%	0.70	0.82
Links	effective date	80%	0.74	0.78
Links	wording	93%	0.84	0.90
Links	new clause id	7%	0.06	0.03
Cases	decision	80%	0.60	0.70
Cases	blocking stage	85%	0.79	0.82
Cases	deciding clauses	15%	0.14	0.11

The new-clause-id field is low mostly because one annotator wrote sub-clauses with a dot (`cl-4.2`) where the registry writes brackets (`cl-4(2)`); the adjudicator resolved these.

S6 Case study: deviations from the analysis plan

All of the following were recorded before any model saw a test item, except the last paragraph.

Amendment 1 (D48).

After the dev pilot, the damaged-registry comparisons (Q1, Q2) replaced the ladder contrasts (R4−R2, R2−R0) as the primary family; the capable models run all 5 damage seeds; discrimination, deference and the looser scoring rules (E5) were added.

Amendment 2 (D49), after the human audit.

- Annotation keeps a case whose adjudicated outcome (decision, and blocking stage for a rejection) matches the key; cases whose deciding clauses alone differ are kept and listed. The planned rule dropped them too and would have kept 6 of 60 cases.
- The correct-answer gate (95% judged correct) and the scorer gate (95% agreement) were replaced by reporting: 57% and 70% were observed. The scorer was not changed.
- Added: the R4n arm and the NoRec contrast (primary family of 10), sensitivity sets S1 and S2, the confusion-count and reference-row reporting rules, descriptive splits, and the registry audit on the random link sample.
- The annotators' meeting before adjudication was dropped; the adjudicator decided every disagreement.
- The analysis cutoff became 24 hours after the local run started, with blocks run in priority order.
- The lead author approved the amendment and stated the co-authors' agreement; not every author signed it.

After the run.

Dated in the decision log. (1) The local models moved from the lead author's laptop to Kaggle GPU notebooks (`qwen2.5:14b`) and back to the laptop for the two smaller models, for speed; each model ran on one machine, and a 30-minute partial start on the laptop that finished no block is kept and unused. Model files were checked against the pre-specified digests on every machine. (2) The analysis cutoff was fixed as a wall-clock time before any test result was read. (3) The laptop's model server crashed once mid-request; the run raised, recorded no reply, and resumed from its cache. (4) Additions after the plan was fixed, outside every planned test: `qwen2.5:32b` (damage seeds 0–2) and Claude Opus 4.5 through a Kaggle-hosted proxy (seed 0) on the main arms, and a same-machine repeat of three blocks to measure determinism. Every planned block finished for the planned models.

S7 Case study: data statement

Curation rationale.

Instruments on rooftop solar (net metering, capacity limits, connection procedures, central subsidy) were collected, including superseded ones, because the study is about replaced rules that remain available.

Language.

English. Six of the 111 documents also contain Hindi passages. A local open model machine-translated these passages into English for retrieval only; the translations are poor in places, and no label depends on them.

Source.

Official publications of central ministries, state electricity regulatory commissions and distribution companies in India, retrieved from their portals. Each document is released as a URL and a hash.

Cases.

Synthetic applications generated from rule functions. No personal data.

Annotators.

Two authors with backgrounds in computing (one an AI engineer with degrees in artificial intelligence and computer science), neither with training in law or energy policy, and a third author adjudicating (§S1.1.1).

Licences.

Code under Apache-2.0; cases and labels under CC BY 4.0; documents as URLs and hashes.

References

Liu Z, Gao S (2026) Trusting the tool: Asymmetric deference of LLM vulnerability auditors to static-analysis evidence. arXiv preprint arXiv:261032876 Submitted to IEEE Access